

AI Teacher

Human-Like AI Educator That Teaches Through Video

Project Documentation — AI Innovation Hackathon 2026, Round 2

1. Problem Statement

Traditional digital learning platforms typically provide either pre-recorded lectures or basic text-based chatbots. Neither approach replicates the personalized, adaptive experience of a real teacher who understands the learner, explains concepts progressively, checks understanding, and adjusts teaching based on how the student responds.

AI Teacher addresses this gap by combining AI-driven lesson generation, spoken narration with a visual avatar, interactive questioning, and adaptive feedback into a single guided learning experience.

2. Solution Overview

AI Teacher is a web application where a student either uploads learning material (a PDF or pasted text) or types a topic they want to learn. The system then:

- Extracts and grounds itself in the uploaded material (or generates structure from the topic alone).
- Generates a structured lesson plan broken into concepts, each with an explanation, a real-world example, and a check-in question.
- Presents the lesson through spoken narration synced with an animated on-screen avatar.
- Evaluates the student's answer to each check-in question and gives targeted feedback, re-explaining when the answer indicates a misunderstanding.
- Concludes with a quiz and a personalized learning report covering score, strong areas, weak areas, and a recommendation for what to revise next.
- Supports multiple languages (English and Hindi in the current build), with the full lesson — explanations, questions, and quiz — generated in the selected language.

3. Key Features

- Material upload with text extraction and an editable preview before lesson generation.
- Topic-based teaching for students without source material.
- Adjustable learner level, explanation depth, concept count, and quiz length.
- Real-time narrated lessons with a synced animated avatar.
- Interactive check-in questions with adaptive, misconception-aware feedback.
- End-of-lesson quiz and personalized learning report.
- Multilingual lesson generation (English, Hindi).
- A pre-recorded AI avatar welcome video introducing the lesson.

4. System Architecture

The application is built as a single web app with the following components:

- Frontend: React-based UI (built and hosted via Lovable), handling material upload, lesson display, narration controls, and the quiz/report flow.
- Backend / data: Supabase, used for storing lesson content, extracted text, and session state.

- Lesson generation: An LLM API call (Claude/OpenAI) that receives the extracted material or topic, along with learner level, time available, and language, and returns a structured lesson (concepts, explanations, examples, check-in questions, and quiz).
- Answer evaluation: The same LLM API is called with the student's response to each check-in question and the target concept, returning feedback that confirms understanding or re-explains the concept differently if the answer suggests a misconception.
- Voice and avatar (per-concept): The browser's native SpeechSynthesis API narrates each concept's explanation aloud in real time, synced with a lightweight animated avatar that pulses while speaking. This works dynamically for any topic at zero cost, since it does not depend on pre-rendering a video per lesson.
- Voice and avatar (intro): A single pre-rendered avatar video, generated using D-ID's avatar video tool, plays as a fixed welcome/introduction to the lesson experience.

5. RAG and Knowledge Grounding

When a student uploads a PDF or pastes text, the system extracts and cleans the text before passing it into the lesson-generation prompt, ensuring the lesson content is grounded in the actual uploaded material rather than the model's general knowledge. The student can preview and edit the extracted text before generation, reducing the risk of the lesson being built on faulty extraction (e.g. from a scanned or poorly formatted document).

For topic-only requests (no uploaded material), the system relies on the LLM's general knowledge to construct the lesson structure, clearly distinguishing this from the material-grounded flow in the UI.

6. Personalization Approach

Before generating a lesson, the student sets:

- Learner level (Beginner / Intermediate / Advanced)
- Explanation depth
- Number of concepts to cover
- Number of quiz questions
- Preferred language

These parameters are passed directly into the lesson-generation prompt, so the depth of explanation, vocabulary, and pacing adjust to the learner rather than producing a one-size-fits-all lesson.

7. Assessment Methodology

Each concept includes a check-in question evaluated immediately by the AI, which responds with feedback rather than a simple right/wrong marker. If the answer reveals a misunderstanding, the AI re-explains the concept before the student moves on. After all concepts are covered, a final quiz assesses overall understanding, and the system compiles a learning report summarizing the student's score, strong areas, weak areas, and a specific recommendation for revision.

8. Multilingual Implementation

The student's selected language is passed as a parameter into every AI call — lesson generation, check-in feedback, and the quiz — so the entire learning experience is delivered in that language, not just the interface labels. The current build supports English and Hindi; the architecture is language-agnostic and can be extended to additional languages by expanding the selectable options.

9. Voice Implementation

Per-concept narration uses the browser's built-in SpeechSynthesis API. This was chosen deliberately over a paid text-to-speech or avatar API so that narration works for any AI-generated topic in real time, at zero cost, rather than being limited to a small number of pre-rendered clips.

10. Avatar / Video Generation Approach

Two complementary approaches are used:

- A lightweight animated avatar (an on-screen circular character) pulses in sync with the browser's speech output during each concept, giving a consistent visual presence throughout the lesson for any topic.
- A single, higher-fidelity avatar video was generated separately using D-ID's avatar video tool and is used as a fixed welcome/introduction clip.

Known limitation: true per-lesson photorealistic avatar video (a unique D-ID/HeyGen-style rendered video for every generated concept) requires paid API access to these services for automated, on-demand generation. This was outside the scope of a free-tier build; the current implementation demonstrates the concept with a real-time animated avatar plus one pre-rendered intro video, and the architecture supports swapping in a full video-generation API in a production deployment.

11. APIs and Third-Party Services Used

- Lovable — application build and hosting platform.
- Supabase — backend data storage.
- Anthropic Claude / OpenAI API — lesson generation, answer evaluation, and quiz/report generation.
- Browser SpeechSynthesis API (native, no external service) — real-time narration.
- D-ID — avatar video generation for the intro clip.

12. Setup Instructions

- The application is deployed and accessible directly at its published URL (see Working Prototype link in the submission).
- No local installation is required to use the app.
- For development: the project is built on Lovable; API keys for the LLM provider are stored securely as environment variables within the platform, not exposed client-side.

13. Deployment Instructions

The application is deployed via Lovable's built-in publishing feature, which builds and hosts the app at a public URL. No separate deployment steps are required for the current build.

14. Known Limitations

- Per-concept avatar video is simulated via real-time browser narration and an animated avatar rather than a photorealistic rendered video, due to the cost of avatar-video APIs at scale.
- The intro avatar video is a single fixed clip rather than dynamically generated per lesson.
- Multilingual support currently covers English and Hindi; additional languages would require expanding the language options and prompt handling.
- The system currently supports single-session lessons; persistent long-term learner profiles across multiple sessions are a planned extension, not yet implemented.

15. Conclusion

AI Teacher demonstrates the core requirement of the challenge: an AI-driven teaching experience that understands material or a topic, plans a lesson, explains it, questions the learner, adapts based on their responses, and assesses their understanding — going beyond a simple question-answering chatbot.

Name a topic or bring your own chapter. AI Teacher turns it into a guided lesson, asks you questions along the way, quizzes you at the end and writes your personal learning report.

What do you want to learn?

e.g. Photosynthesis, Newton's laws, React hooks

Study material (optional)

Paste your notes, chapter text or transcript here...

Upload a PDF or text file 0 words · 0 / 40,000 characters

Lesson settings

Learner level

Explanation depth

Concepts 4

Quiz questions 5

Lesson language

Explanations, questions and the quiz are all written in this language.

Create my lesson